



Data Analysis

*Project 2: Travel, Tourism & Hospitality -
Customer Retention and Dynamic Pricing
Analysis*

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Introduction



This project analyzes hotel booking data to understand customer behavior and identify key factors influencing booking cancellations.

The goal is to support better pricing strategies and improve customer retention in the hospitality industry.





Problem

Statement



In the highly competitive travel and hospitality sector, agencies and hotel chains suffer from significant revenue leakage due to unoptimized pricing and unpredictable customer cancellations. Traditional booking systems lack the predictive intelligence to anticipate when a user is likely to cancel a reservation or to adjust room rates dynamically based on fluctuating demand and seasonality.

Dataset Description

The dataset contains hotel booking records for both City Hotels and Resort Hotels, capturing customer behavior and booking patterns.

It includes features such as:

- Lead Time (advance booking period)
- Hotel Type (City / Resort)
- Market Segment (booking source)
- ADR (Average Daily Rate)
- Customer Type
- Deposit Type
- Total Nights Stayed
- Special Requests

The target variable is `is_canceled`, which indicates whether a booking was canceled or not.



Data Cleaning



Handled missing values

missing values in columns like children and country



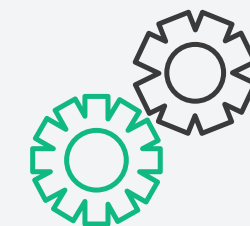
Removed invalid records

Removed invalid records (e.g., adults = 0, ADR < 0)



Checked duplicates

Checked and handled duplicate records carefully



Ensured correct data types

Ensured correct data types for numerical and date columns

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify patterns, trends, and key factors influencing booking cancellations and customer behavior

Key analyses performed:

- Lead time vs cancellation
- ADR vs cancellation
- Market segment vs cancellation
- Deposit type vs cancellation
- Seasonal booking trends

These analyses helped uncover important relationships between booking behavior, pricing, and cancellation risk.



Feature Engineering

New features created to improve analysis:



Total Guests

Sum of adults, children, and babies



Total Nights

Total duration of stay
(weekend + weekday)



Booking Season

Total duration of stay
(weekend + weekday)



Lead Time Category

Grouped into booking
behavior segments

Model Building

Machine learning models were developed to predict whether a hotel booking will be canceled based on historical booking data.

Models used:

- Logistic Regression
- Random Forest

These models were trained using key features such as lead time, market segment, deposit type, ADR, and customer behavior.

The objective was to identify high-risk bookings and support proactive decision-making.



RESULTS

Logistic Regression: ~78% Accuracy

Random Forest: ~85% Accuracy

Random Forest performed better in predicting cancellations

Model EVALUATION



The performance of the models was evaluated using key classification metrics to measure prediction accuracy and reliability.

Evaluation metrics used:

- Accuracy – Overall correctness of predictions
- Precision – Correctly predicted cancellations out of total predicted
- Recall – Ability to identify actual cancellations
- F1 Score – Balance between precision and recall
- Confusion Matrix – Comparison of actual vs predicted outcomes

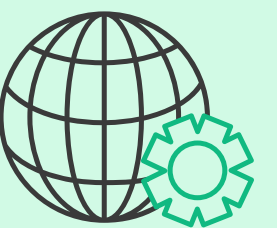


Dashboard (Power BI)

An interactive dashboard was developed to visualize booking trends, customer behavior, and cancellation patterns.

Key features:

- KPI cards for quick overview (Revenue, Bookings, Cancellation Rate, ADR)
- Analysis of customer behavior and booking patterns
- Cancellation analysis based on lead time, deposit type, and market segment
- Pricing insights using ADR
- Seasonal trends and demand patterns
- Interactive slicers for dynamic filtering



Hotel Booking Analytics: Customer Retention & Dynamic Pricing

Booking Overview & Customer Behavior Insights

37.08%

Cancellation Rate%

42.65M

Total Revenue

119K

Total bookings

104.07

Average Lead Time

102.01

Average ADR

HOTEL

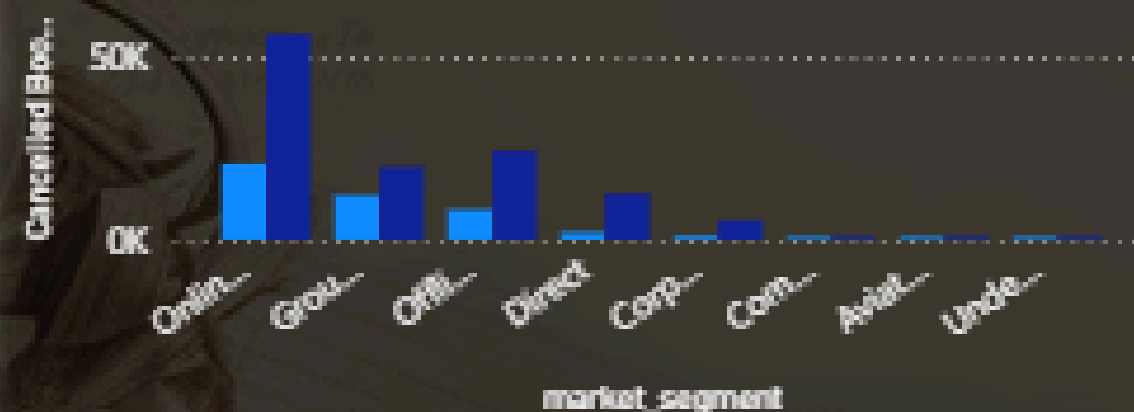
- ☒ City Hotel
- ☐ Resort Hotel

market_segment

- ☐ Select all
- ☐ Aviation
- ☐ Complementary
- ☐ Corporate
- ☐ Direct

Cancelled vs Total Bookings by Market Segment

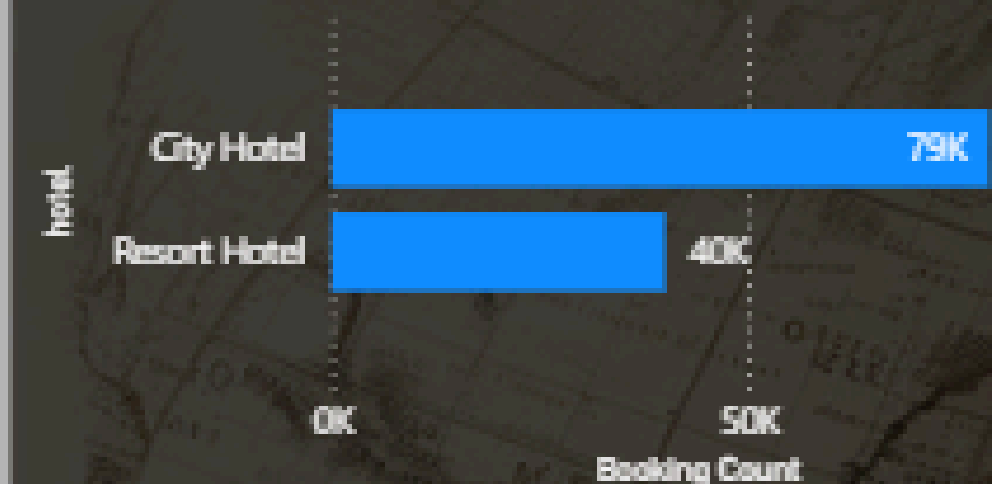
Cancelled Bookings Total bookings



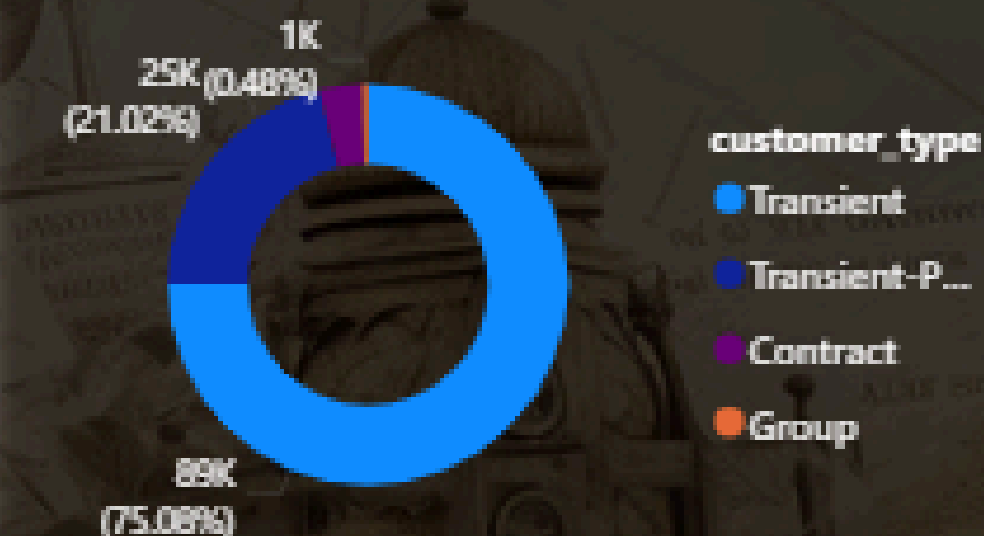
Repeat vs New Guest Bookings



Total Bookings by Hotel Type



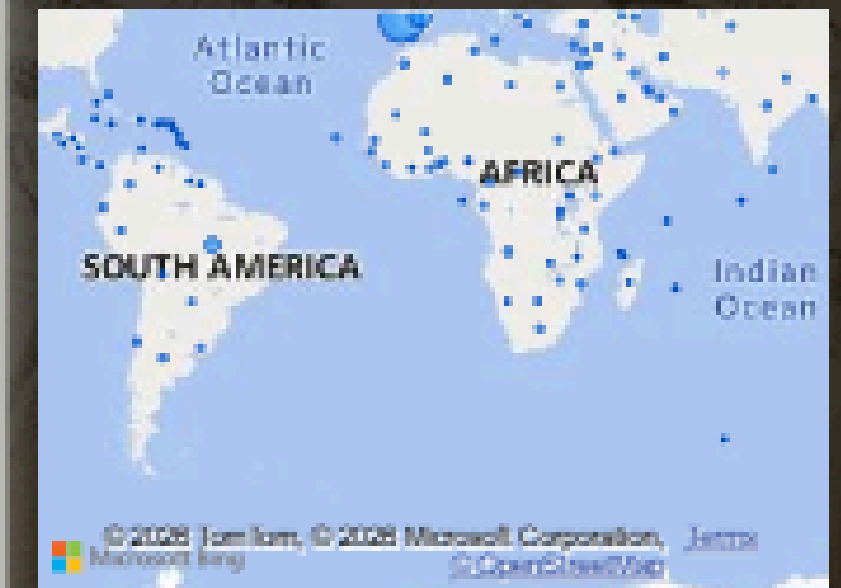
Bookings by Customer Type



Bookings by Total Nights Stayed



Booking Distribution by Country



KEY INSIGHTS

The analysis revealed key factors influencing booking cancellations and customer behavior

- Higher lead time → Higher cancellation probability
- No-deposit bookings → Highest cancellation risk
- Online travel agents → High bookings but lower reliability
- More special requests → Lower cancellation likelihood
- Seasonal trends → Strong impact on demand and pricing



Conclusion & Business Impact



Key outcomes:

- Identified major drivers of cancellations
- Developed a predictive model for cancellation risk
- Provided insights for dynamic pricing strategies
- Highlighted opportunities to improve customer retention

Business Impact:

- Helps reduce revenue loss due to cancellations
- Enables better pricing decisions based on demand
- Supports targeted retention strategies
- Improves overall operational efficiency



Thank You

